CENTRIFUGAL PUMP

The hydraulic machine which converts the Mechanical energy into Hydraulic Energy is called Pumps. The hydraulic energy is in the form of pressure energy.

PUMPS CLASSIFICATION:

On the basis of transfer of mechanical energy, pumps are classified as following categories:

I. **Rotodynamic pumps:**
   a) Radial flow pumps
   b) Axial flow pumps
   c) Mixed flow pumps

II. **Positive displacement pumps**

The **radial flow type pumps** are commonly called **Centrifugal pumps.** The energy transfer in a radial flow type pumps occurs mainly when the flow is in the radial path.

ADVANTAGES OF CENTRIFUGAL PUMP OVER POSITIVE DISPLACEMENT PUMP (RECIPROCATING PUMP):

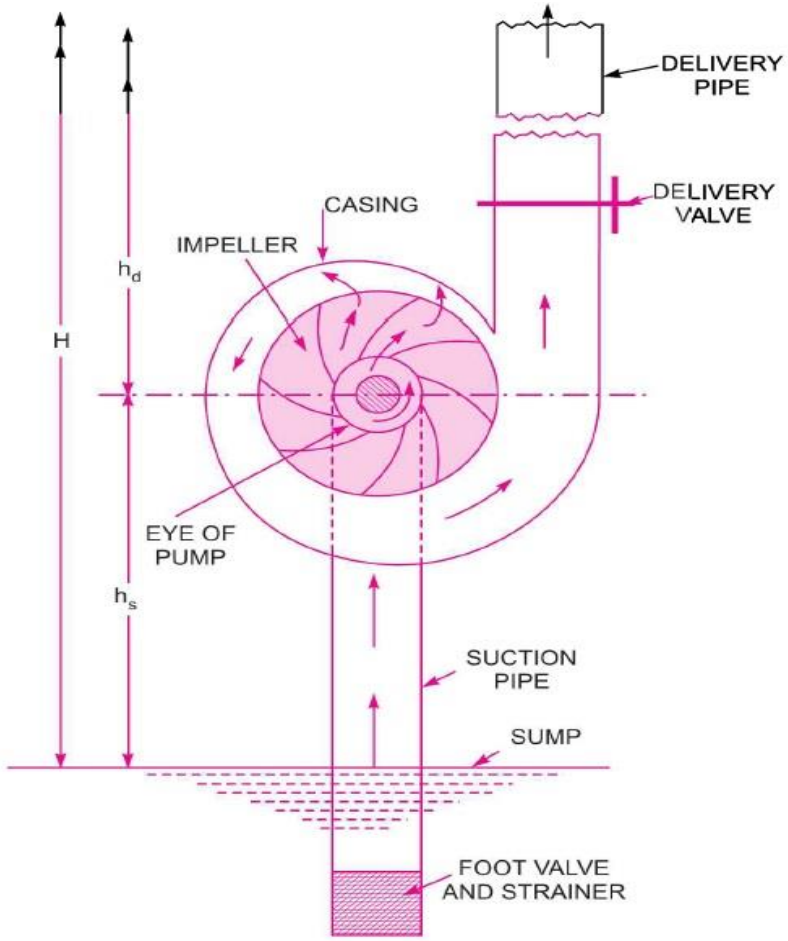
The centrifugal pump claims the following advantages with reference to a positive displacement (reciprocating) pump:

1. The cost of a centrifugal pump is less as it has fewer parts.
2. Installation and maintenance are easier and cheaper.
3. Its discharging capacity is much greater than that of a reciprocating pump.
4. It is compact and has smaller size and weight for the same capacity and energy transfer.
5. Its performance characteristics are superior.
6. It can be employed for lifting highly viscous liquid such as paper pulp, muddy and sewage water, oil, sugar molasses etc.
7. It can be operated at very high speeds without any danger of separation and cavitation.
8. It can be directly coupled to an electric motor or an oil engine.
9. The torque on the power source is uniform, the output of the pump is also uniform.

COMPONENT PARTS OF A CENTRIFUGAL PUMP:

A centrifugal pump consists of the following main components:

1. Impeller 2. Casing 3. Suction pipe with a foot valve and strainer 4. Delivery pipe.



Impeller:

It is a wheel or rotor which is provided with a series of backward curved blades or vanes. It is mounded on the shaft which is coupled to an external source of energy which imparts the liquid energy to the impeller there by making it to rotate.

Impellers are divided into 3 types,

1. Open Impeller
2. Semi enclosed Impeller
3. Enclosed Impeller

Casing:

It is a pipe which is connected at the upper end to the inlet of the pump to the centre of impeller which is commonly known as eye. The double end reaction pump consists of two suction pipe connected to the eye from both sides. The lower end dips into liquid in to lift. The lower end is fitted in to foot valve and strainer.

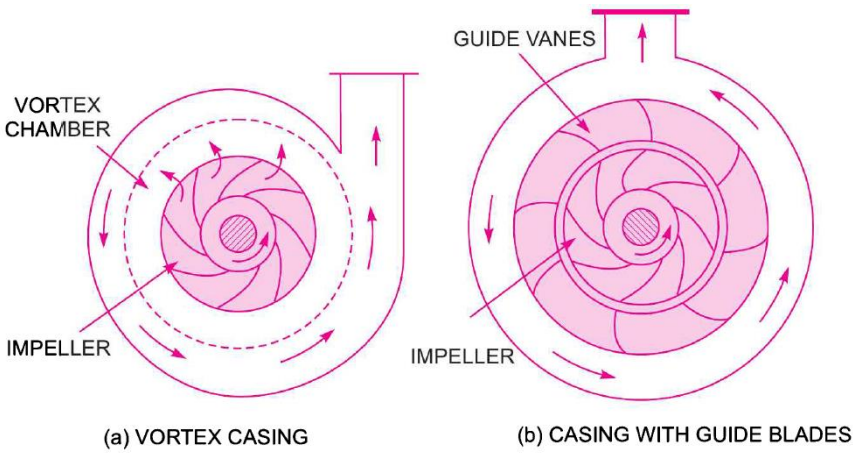
Commonly three types of casing are used in centrifugal pump,

1. Volute Casing
2. Vortex Casing
3. Casing with Guide Blades

Volute casing: It is a spiral type in which area of flow increase gradually. The increase in area of flow decreases the velocity of flow. Decrease in velocity increases pressure of water flowing

Vortex casing: Vortex casing is a casing in which circular chamber is provided between the casing and the impeller. Vortex casing is helpful to increase pump efficiency by reducing eddies formation to a considerable extent through the casing.

Casing with Guided Blades: In the casing with guided blades the impeller is surrounded by series of guide blades. The guide blades are mounted on a ring which is called as diffuser. The design of the guide vanes are kept as such that the water which is leaving the impeller enters the guides without shock. The area of the guide vanes increases; this helps to decrease the velocity of the liquid and increases its pressure. After guide vanes, water passes through the surrounding casing. In most of the cases, the casing remains concentric with the impeller.



(a) VORTEX CASING (b) CASING WITH GUIDE BLADES

Delivery Pipe:

It is a pipe which is connected at its lower end to the out let of the pump and it delivers the liquid to the required height. Near the outlet of the pump on the delivery pipe, a valve is provided which controls the flow from the pump into delivery pipe.

Suction Pipe with Foot Valve and Strainer:

suction pipe is connected with the inlet of the impeller and the other end is dipped into the sump of water. At the water end, it consists of foot value and strainer. The foot valve is a one way valve that opens in the upward direction. The strainer is used to filter the unwanted particle present in the water to prevent the centrifugal pump from blockage.

WORKING OF A CENTRIFUGAL PUMP:

A centrifugal pump works on the principle that when a certain mass of fluid is rotated by an external source, it is thrown away from the central axis of rotation and a centrifugal head is impressed which enables it to rise to a higher level.

The working /operation of a centrifugal pump is explained step-wise below:

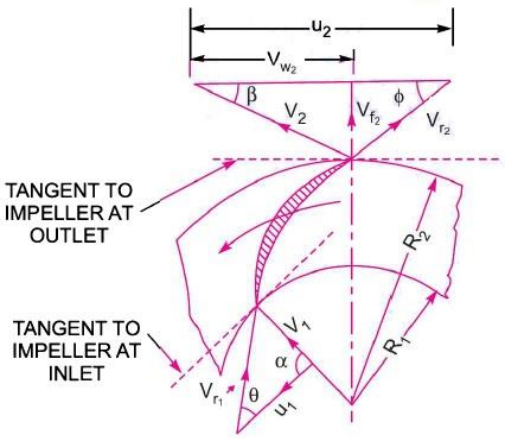
1. The delivery valve is closed and the pump is primed that is, suction pipe, casing and portion of the delivery pipe upto the delivery valve are completely filled with the liquid (to be pumped) so that no air pocket is left.
2. Keeping the delivery valve still closed the electric motor is started to rotate the impeller. The rotation of the impeller causes strong suction or vaccum just at the eye of the casing.
3. The speed of the impeller is gradually increased till the impeller rotates at its normal speed and develops normal energy required for pumping the liquid.
4. After the impeller attains the normal speed, the delivery valve is opened when the liquid is continuously sucked (from sump well) up the suction pipe, it passes through the eye of casing and enters the impeller at its centre or it enters the impeller vanes at their inlet tips. This liquid is impelled out by the rotating vanes and it comes out at the outlet tips of the vanes into the casing.

Due to impeller action the pressure head as well as velocity heads of the liquid are increased (some of this velocity heads is converted into pressure head in the casing and in the diffuser blades/vanes if they are also provided).

5. From casing, the liquid passes into pipe and is lifted to the required height (and discharged from the outlet or upper end of the delivery pipe).
6. So long as motion is given to the impeller and there is supply of liquid to be lifted the process of lifting the liquid to the required height remains continuous.
7. When pump is to be stopped the delivery valve should be first closed, otherwise there may be some backflow from the reservoir.

WORK DONE BY THE IMPELLER (OR CENTRIFUGAL PUMP) ON LIQUID:

In case of centrifugal pump, work will be done by the impeller on the water. Expression for the work done by the impeller on the water will be determined by drawing the velocity triangles at inlet and outlet of the impeller.
For best efficiency of the pump, water need to enter the impeller radially at inlet. Therefore, absolute velocity of water at inlet will make an angle of 90 degree with the direction of motion of the impeller at inlet. Therefore, angle $\alpha = 90^0$ and velocity of whirl at inlet $V_{w1} = 0$.



Work done by the impeller on water per second = (W/g) x V_{w2} u_2

Work done by impeller to the water per unit weight per second = (1/g) x V_{w2} u_2

HEAD OF A CENTRIFUGAL PUMP:

The head of a centrifugal pump may be expressed in the following ways:

Static head; (ii) Manometric head; (iii) Total, gross or effective head

I. **Static head** is basically defined as the sum of suction head and delivery head and it will be displayed by the symbol H_S. Mathematically we will write the expression for static head as $\mathbf{H_S = h_s + h_d}$

II. **Manometric head** is basically defined as the head against which a centrifugal pump has to do the work. Manometric head will be displayed by H_m.

Manometric head = Head imparted by the impeller to the water – loss of head inside the pump

Manometric head = $(V_{w2}\, u_2)/g$ - loss of head inside the pump

If there is no loss of head inside the pump, we will have following expression for manometric head

$$\mathbf{Manometric\ head = (V_{w2}\, u_2)/g}$$

Where,

V_{w2} = Velocity of whirl at outlet

u_2 = Tangential velocity of impeller at outlet

g = Acceleration due to gravity

Manometric head could be expressed by the following expression also.

Manometric head = Suction head + Delivery Head + Frictional head loss in suction pipe + Frictional head loss in discharge pipe + Velocity head in delivery or discharge pipe

$$\mathbf{Manometric\ head = h_s + h_d + h_{fs} + h_{fd} + V_d^2/2g}$$

III. **Total head** is equal to the static head plus all the head losses occurring in flow before, through and after the impeller.

Efficiencies of a centrifugal pump:

Power will be transmitted from electrical motor shaft to shaft of the centrifugal pump. Further power will be transferred from shaft of the centrifugal pump to impeller of the centrifugal pump. Further, power will be transferred from pump impeller to the water.

There will be some losses of power when power will be transmitted between pump shaft to pump impeller and further between pump impeller to the water.

There will be following efficiencies related with centrifugal pump:

1. Manometric Efficiency, η_{man}
2. Mechanical Efficiency, η_m
3. Overall efficiency, η_o

Manometric Efficiency, η_{man} :

Manometric Efficiency is defined as the ratio of manometric head to the head imparted by centrifugal pump impeller to the water. Mathematically,

Manometric efficiency, η_{man} = Manometric Head / Head imparted by centrifugal pump impeller to the water

The power at the impeller of the pump will be more than the power given to the water at outlet of the pump. Manometric efficiency will also be defined as the power given to the water at outlet of the pump to the power available at the impeller of the pump.

Manometric efficiency, η_{man} = Power given to the water at outlet of the pump / Power available at the impeller of the pump

$$\eta_{man} = \frac{g \times H_m}{V_{w_2} \times u_2}$$

Mechanical efficiency:

Power at the shaft of the centrifugal pump will be more than the power available at the pump impeller. Mechanical efficiency will be defined as the ratio of power available at the pump impeller to the power available at the shaft of the centrifugal pump.

Mechanical efficiency, η_{mech} = Power available at the impeller of the pump/ Power available at the shaft of the centrifugal pump

$$\eta_m = \frac{\frac{W}{g}\left(\frac{V_{w_2} u_2}{1000}\right)}{\text{S.P.}}$$

Overall efficiency:

Overall efficiency will be defined as the ratio of power output of the pump to the power input to the pump.

Power output of the pump in KW = Weight of water lifted (W) x H_m /1000

Power input to the pump = Power supplied by the electric motor = S.P.

$$\eta_o = \frac{\left(\frac{WH_m}{1000}\right)}{\text{S.P.}}$$

$$\eta_o = \eta_{man} \times \eta_m.$$

MULTI-STAGE CENTRIFUGAL PUMPS:

A multi-stage centrifugal pump is one which has two or more identical impellers mounted on the same shaft or on different shafts. The important functions performed by a multi-stage pump are:

1. To produce heads greater than that permissible with a single impeller, 'discharge remaining constant'. The task can be achieved by 'series arrangement' where in the impellers are mounted on the same shaft and enclosed in the same casing.

2. To discharge a large quantity of liquid, 'head remaining same'. This task is accomplished by 'parallel arrangement' wherein impellers are mounted on separate shafts.

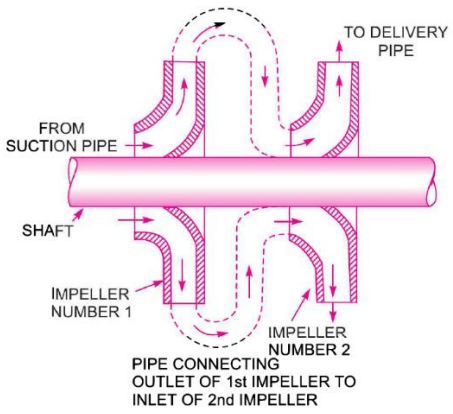


Fig: Impeller in series arrangement

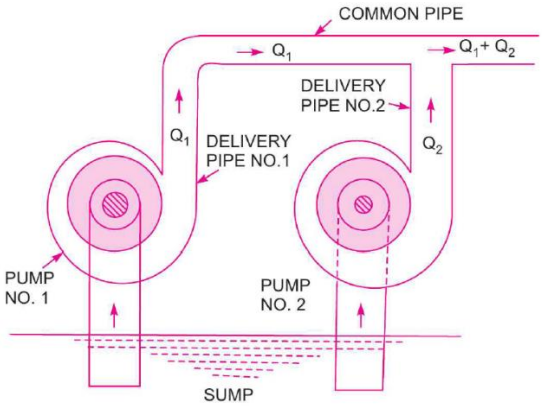


Fig: Impeller in parallel arrangement

PRIMING:

Pump Priming is the process of removing air from the pump and suction line. In this process the pump is been filled with the liquid being pumped and liquid forces all the air, gas, or vapor contained in the passage ways of pump to escape out. Priming maybe done manually or automatically. Priming a pump is the first and one of the most important operation one should do before operating it. Priming reduces the risk of pump damage during start-up as it prevents the pump impeller to becomes gas-bound and thus incapable of pumping the desired liquid. For reliable operation, pumps must first be primed; that is, air or gases to be expelled from the suction and impeller eye area and replaced with liquid to be pumped. The pump would not function properly when not completely filled with liquid. Along with compromised performance, not priming the pump and allowed to run without fluid, it will overheat the pump system and there will be a danger of damage to critical internal pump components.

SPECIFIC SPEED:

The **specific speed** of a centrifugal pump is defined as the speed of a geometrically similar pump which would deliver unit quantity (one cubic metre of liquid per second) against a unit head (one metre). It is denoted by Ns. The specific speed is a characteristic of pumps which can be used as a basis for comparing the performance of different pumps.

$$Ns = N\,Q^{1/2} / (H_{mano})^{3/4}$$